

Mazi Umntanakho “Know Your Child”: An Accessible Social-Emotional Assessment Tool for Children in Low-Income South African Communities

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ABSTRACT

Ensuring proper developmental and behavioral assessments for children is crucial, particularly in contexts where resources are scarce, developmental delays carry a stigma, and specialists are lacking. However, adequate and culturally appropriate methods for screening and assessing children in the Majority World still need to be improved. In this context, mobile technology can be crucial in providing accessible assessments and screenings that better fulfill the needs. In this work, we present the design and development of Mazi Umntanakho “Know Your Child,” a conversational agent on WhatsApp that supports South African home visitors in assessing and tracking children’s socio-emotional skills. The agent is based on the Strengths and Difficulties Questionnaire (SDQ) and International Development and Early Learning Assessment (IDELA) to provide culturally appropriate assessments. The agent is personalized to user preferences and language and provides feedback and advice in various multimedia formats, including infographics, video, audio, and photos.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing.**

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KEYWORDS

Mobile health (mHealth), conversational agent, early childhood development, culturally appropriate methods

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1 INTRODUCTION

Ensuring the proper assessment and screening of development in early childhood is crucial as early childhood is characterized by a rapid period of neurodevelopment, particularly in the development of executive functions (EF) [4, 5]. EF plays a vital role in mental health as it contributes to self-regulation [10, 14] and predicts academic achievement, school readiness, lifelong achievement, health, wealth, and quality of life [3, 6].

Particularly in contexts with scarce resources, such as low- and middle-income countries (LMICs), the proper assessment of developmental milestones, especially EF, can make a huge impact on children and their families. Unfortunately, in LMICs, developmental delays often carry a stigma, specialists are lacking, and the symptoms of neurodevelopmental delays may be misunderstood or misconstrued in [17]. Particularly in South Africa, there is an urgent need for developmental screening at a community level to ensure

that neurodevelopmental delays are identified early so children in need of intervention have more chances to receive timely support.

However, adequate and culturally appropriate methods for screening and assessing children in the Majority World¹ still need to be improved. In these contexts, mobile health (mHealth) could play a crucial role by shortening the gap and providing accessible assessments and screenings that better fulfill the needs of these populations. Literature on Ubiquitous technology and Human-Computer Interaction has shown the potential of mHealth to address health inequalities when they are properly designed [8, 9, 12]. For instance, in South Africa, MomConnect, a text messaging service, provides health information to pregnant and postpartum women [13, 15], showing the potential of ubiquitous and mobile technology to support families even when resources are scarce.

2 DESIGN METHODS

To understand the needs of the home visitors, four members of the research team conducted 19 focus groups across two weeks on May 22. These groups included 51 home visitors and were focused on understanding the perceptions of technological tools to provide assessments and evaluations in low-income communities across South Africa. The focus groups were conducted in both urban ($n = 6$ groups, 34 people) and rural ($n = 4$ groups, 17 people) communities near Cape Town, South Africa. Each session lasted about 60 minutes.

We then designed and developed the prototype Mazi Umntanakho WhatsApp conversational agent. Following, we sought feedback from a different set of home visitors using videos of the tool as well as other design artifacts. Two of four researchers who conducted the first set of focus groups then conducted a second round of focus groups ($n = 8$ groups) in October 2022 focused on collecting feedback about the WhatsApp conversational agent.

The empirical data collected during these interviews is analyzed and presented elsewhere. The findings that relate directly to the design of the Mazi tool and its requirements and development of it are the focus of this work. In the next subsection, we outline the design requirements and the development of the proposed conversational agent.

2.1 Mazi Umntanakho Conversational Agent

The overarching project that this work is a part of seeks to design, develop, and evaluate a mHealth screening tool for home visitors at LMICs to 1) screen for developmental delays and assess executive function development in young children (ages 2 to 5 years) in low-income settings in South Africa and 2) provide a community-based treatment paradigm. To accomplish the first aim, building on feedback from the home visitor, we develop a tool aimed at providing comprehensive support to visitors during assessments and evaluations.

To provide culturally appropriate assessments and evaluations, the Mazi Umntanakho conversational agent includes simplified versions of assessments used in LMICs translated into multiple languages (see Figure 1). At the end of each assessment, to provide support about the outcomes, the agent automatically computes the score and provides real-time feedback in different formats that can

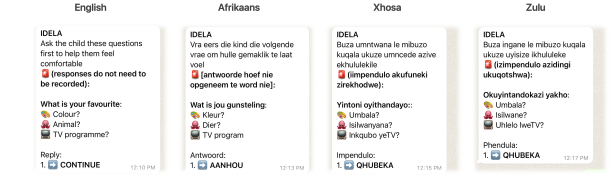


Figure 1: Assessments used in LMICs translated into multiple languages.

be easily shared and understandable among the stakeholders. Moreover, it retains the information over time for repeat assessments in case home visitors conduct a follow-up of particular children. The following subsections highlight the key design insights that guided our development efforts.

2.1.1 Assessments. In this work, first, we focused on the use of validated culturally appropriate assessments of social and emotional development, ultimately choosing The Strengths and Difficulties Questionnaire (SDQ) [7] and the social-emotional component of the International Development and Early Learning Assessment (IDELA) [16]. These tools were selected due to their use across diverse settings, their ability to be translated into local South African languages (with some translations already available), the feasibility of administration by community-based workers, the salience of the outcomes assessed, and their potential acceptability in the contexts where the project was being conducted [11].

The SDQ, widely used globally to assess mental health symptoms in children, is a brief emotional and behavioral screening tool typically administered by a caregiver. On the other hand, children perform a set of activities so home visitors can score four domains of the IDELA motor, literacy, numeracy, and social-emotional. In this work, we only used the social-emotional component, which include 16 questions focused on self-awareness, friends, emotional awareness, empathy, and conflict resolution. This tool, developed by Save the Children, is intended for children 3 to 6 years of age and was intentionally developed for a global and frequently low-income or low-education environment².

As a result of the focus group, a simplified version of SDQ and IDELA socio-emotional components were carefully designed by two members of the research team experts in child development to make sure that they are understandable by the home visitors and families and they still provide enough information to evaluate socio-emotional development in this context. The Mazi Umntanakho conversational agent consists then in providing a set of questions that the home visitors will ask the parent (SDQ) or the child (IDELA) and they will enter the information into the tool for scoring.

2.1.2 Multiple languages. South Africa has eleven official languages, including English, Afrikaans, and nine ethnic languages [1], with Zulu (22.7%) and Xhosa (16%) being the most prevalent. Thus, our agent needed to address the country's linguistic diversity with multi-language capabilities to enable users to access and utilize Mazi's resources and functionalities in their preferred language. While not a replacement for home visitors who must also be multi-lingual, provision of support through all possible languages of the

¹Majority World refers to the collective group of countries that together comprise the majority of the global population [2].

²<https://idela-network.org/about/countries-where-idela-is-in-use/>

region enables individuals across different communities to benefit from its support. Currently, the chatbot facilitates communication in English, Afrikaans, Xhosa, and Zulu. However, the upcoming second phase is expected to include North Sotho, South Sotho, Tsonga, Tswana, and Venda.

2.1.3 Real-Time Scoring & Feedback. The simplified version of the SDQ consists of 20 questions distributed across five sub-scales: emotional challenges, conduct problems, hyperactivity, peer relationship problem, and prosocial behavior. Each question in SDQ is assigned a value from 0 to 2. The score for each sub-scale can be seen in Figure 2.1. On the other hand, IDELA consists of 16 questions across five sub-scales: self-awareness, Friends, emotional awareness, empathy, and solving conflicts. Each question in IDELA is assigned a value from 0 to 1.

In the second focus group we found that a numeric score may not be feasible to provide feedback to the families, therefore, to make the scoring easier for the home visitors and families, we propose a simplified approach mapping the numeric scores to colors based on the robot light system (*i.e.*, traffic light), where red indicates scores that require attention in a specific domain and subscales, orange indicates in what domains the child is having a few challenges, so the caregiver needs to keep an eye on these domains, and green means that children may not exhibit challenge in those domains.

Using a color-coded visual representation makes it easier, more accessible, and more intuitive for the caregiver to understand which domains need further intervention (see Figure 2.1).

The Mazi Umntanakho conversational agent provides feedback as soon as the home visitor finishes the questions from the assessment for each of the domains and as a whole. For the simplified version of SDQ, the chatbot selects a color for each of the five sub-scales evaluated 1) Emotional challenges, 2) conduct problems; 3) Hyperactivity/Inattention; 4) Peer relationship problems; 5) prosocial behavior. For the simplified version of IDELA socio-emotional assessment, the chatbot provided one color related to the social-emotional domain (see Figure 2.2). If more than one domain requires attention, the agent automatically provides this information to the home visitor. The Mazi Umntanakho conversational agent also evaluates and suggests the home visitors speak to their supervisor if a particular child satisfies the following criteria: (1) The child's red score for any of the domains; (2) The child's caregiver has many concerns; (3) The home visitor has many concerns.

2.1.4 Access Information Anytime for Everyone. In order to ensure equitable and inclusive access to mental health information, it is crucial to align the preferences and abilities of caregivers. Given the low literacy levels of the families that the home visitors will visit to make the assessment, we have identified the need to present information in multiple formats, including photos, text, videos, and voice messages.

For the Mazi Umntanakho, a text, infographic, audio, and video were created for each domain: self-awareness, emotions, social skills, difficult behavior, health habits, and routines. Figure 3 shows information related to emotions in the different multimedia content. The information is provided on-demand if the parents request it. That multimedia information provides caregivers with valuable information and tips concerning their child's mental well-being.

2.1.5 Long-term Tracking and Repeat Assessments. The SDQ and IDELA assessments are crucial in tracking a child's developmental progress. An initial assessment is essential to establish a reference point for measuring development and intervention effectiveness over time. Mazi Umntanakho conversational agent can collect this initial assessment, enabling a comprehensive understanding of a child's progress and the impact of interventions.

Focus group feedback showed that different home visitors could assess children over time, and health services organizations should have mechanisms that allow them to track each child's progress. Moreover, it is essential to prioritize protecting the security and privacy data of families and their children.

The Mazi Umntanakho incorporates features that enable the collection and storage of assessment data securely and privately. Each child is assigned an encrypted ID code, allowing tracking multiple instances of the assessment and protecting individually identifiable health information.

3 IMPLEMENTATION

Figure 4 shows a high-level integration workflow for Mazi Umntanakho. Whatsapp serves as the primary communication platform, enabling end-user interaction. To communicate between the Mazi tool and WhatsApp, we use Twilio. On the other hand, we used Flask and Python to develop the chatbot logic and functionalities. To store and manage the relevant data for the chatbot application, we used Heroku PostgreSQL. Finally, we used Heroku as the cloud hosting platform to make a scalable, reliable tool.

3.1 The Mazi Umntanakho Conversational Agent Scenario

In the following scenario, we exemplify the use of The Mazi Umntanakho:

Nontando is a home visitor who lives in Cape Town and speaks Xhosa. Today she will visit Milani, a 5-year-old boy, and his family to assess his emotional development. She brings her phone with WhatsApp installed. Upon arriving, she greets his mom and sends a message to the Mazi Umntanakho via WhatsApp. The conversational agent responds: "Hello! Welcome to the Mazi Umntanakho. Please enter the child's unique ID" Nontando introduces Milani's ID, and then it requests the language preference, and she selects Xhosa. Then the agent asks Milani's mom's consent to conduct the assessment. Next, the agent asks for the age of Milani in years and Nontando type 5. The agent then asks "Which questions would you like to start with? Reply: 1. SDQ; 2. IDELA" Nontando chooses IDELA by typing 2. Then the agent starts presenting the questions individually, and Nontando type 1 if the child answers correctly or 2 if he refuses. For example, the conversational agent presents, "What is your name and surname? Response options: 1. Correct; 2. Incorrect or do not know". After answering the 16 questions, the agent displays a green robot traffic light. Nontando reads the message to Milani's mom "The assessment shows that your child is not having any challenges with their social and emotional development." Then Nontando ask Milani's mom if she would like tips for any of the domains presented in the agent, and she answers that for emotions. So Nontando asks in what format, and the Milani's mom answers via video. Then the agent shows a video in Xhosa to explain more about emotional development. After

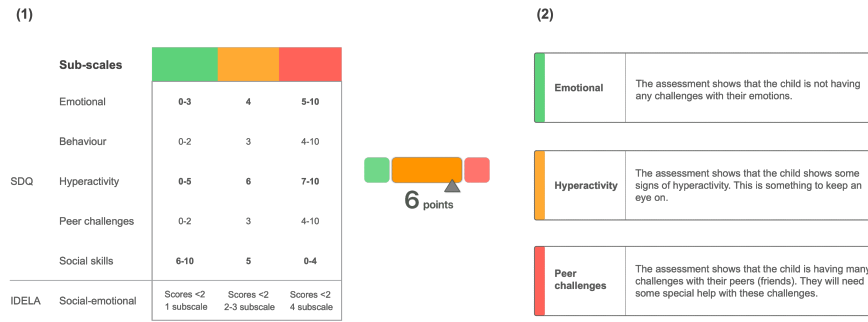


Figure 2: Color-coded visual representation for SDQ and IDELA score results. (1) Employing a traffic light system to map numeric scores to color codes. (2) Example of feedback to support the children's mental health development.

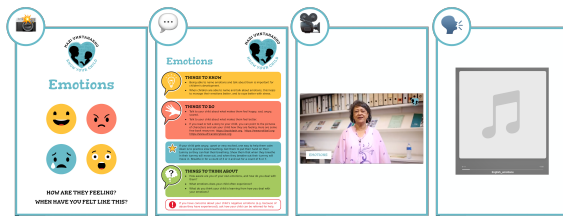


Figure 3: The agent is personalized to user preferences and language, giving information and advice in various formats such as photos, text, video, and audio.

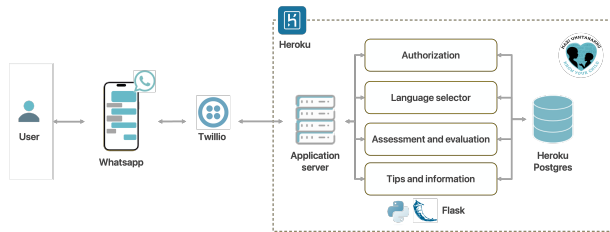


Figure 4: A high-level integration workflow for Mazi Umntanakho.

showing the video, Nontando says good by to Milani and his mom, and she concludes the assessment successfully.

4 CURRENT AND FUTURE WORK

The Mazi Umntanakho conversational agent is currently in the pilot phase, which aims to test the usability and acceptability of the agent among low-income families in South Africa. During this phase, the agent was integrated into different communities, and activities such as user training and technical support were provided to ensure successful implementation. The pilot phase will last at least six months. Currently, we have 144 home visitors and 4243 children registered. From those, 75 home visitors have successfully used the agent to evaluate 434 children. At the end of the six months, we will conduct a third session of focus groups to determine whether the intervention was helpful for caregivers and improved outcomes for children.

5 CONCLUSION AND DISCUSSION

The initiative to support home visitors in screening and tracking children's social and emotional skills through WhatsApp is a positive step towards improving child development outcomes. WhatsApp's popularity in South Africa makes it a ubiquitous and extensible platform for facilitating effective communication and collaboration among home visitors, caregivers, and healthcare organizations. However, it is crucial to consider the specific needs of the communities to ensure accessibility, usability, and understanding for all users. The proposed technology provides clear and timely feedback, allowing non-specialized individuals to conduct the assessment and advise the families.

Existing assessment tools for socio-emotional development, evaluated in different contexts, are usually paper-based or require downloading a specific mobile application. In addition, these assessments are typically available in mainstream languages, using wording and jargon that mainly experts can understand. Therefore, an essential aspect of ensuring accessibility in assessment is to employ simplified grammar and commonly used languages within the communities to break down the language barriers. Moreover, leveraging technology becomes crucial in contexts with limited access to specialists. In this project, we developed a conversational agent with real-time feedback to simplify assessments and enable home visitors to conduct them more widely. Translations and simplified versions of the assessment were also provided to enhance usability.

Overall, mHealth tools allow patient self-interaction and provide feedback to users anytime, reducing reliance on clinicians for surveys and results. Future work should focus on evaluating the effectiveness of this intervention to determine its impact on home visitors, caregivers, children, and organizations. A qualitative evaluation after six months will provide insights into the strengths and weaknesses of the intervention and identify areas that need improvement.

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